

ITC3154 – Cognitive Computing – Assignment 1 Report

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GitHub of assignment: <https://github.com/FokionKonstantaras/ITC3154-Assignment-1/tree/main>

Task 2:

input												
loan_id	no_of_dependents	education	self_employed	income_annum	loan_amount	loan_term	cibil_score	residential_assets_value	commercial_assets_value	luxury_assets_value	bank_asset_value	loan_status
1	2	Graduate	No	9600000	29900000	12	778	2400000	17600000	22700000	8000000	Approved
2	0	Not Graduate	Yes	4100000	12200000	8	417	2100000	2200000	8800000	3200000	Rejected
3	3	Graduate	No	9100000	29700000	20	506	7100000	4500000	33300000	12800000	Rejected
4	3	Graduate	No	8200000	30700000	8	467	18200000	3300000	23300000	7900000	Rejected
5	5	Not Graduate	Yes	9800000	24200000	20	382	12400000	8200000	29400000	5000000	Rejected
6	0	Graduate	Yes	4800000	11500000	10	319	4800000	8300000	13700000	3100000	Rejected
7	5	Graduate	No	8700000	33000000	4	678	22500000	14800000	29200000	4300000	Approved
8	2	Graduate	Yes	5700000	15000000	20	382	13200000	5700000	11800000	6000000	Rejected
9	0	Graduate	Yes	800000	2200000	20	782	1300000	800000	2800000	600000	Approved
10	5	Not Graduate	No	1100000	4300000	10	388	3200000	1400000	3300000	1600000	Rejected
11	4	Graduate	Yes	2900000	11200000	2	547	8100000	4700000	9500000	3100000	Approved
12	2	Not Graduate	Yes	6700000	22700000	18	538	15300000	3800000	20400000	6400000	Rejected
13	3	Not Graduate	Yes	5000000	11600000	16	311	6400000	9600000	14600000	4300000	Rejected
14	2	Graduate	Yes	9100000	31500000	14	679	10800000	16600000	20900000	5000000	Approved
15	1	Not Graduate	No	1900000	7400000	6	469	1900000	1200000	5900000	1900000	Rejected
16	5	Not Graduate	No	4700000	10700000	10	794	5700000	3900000	16400000	4400000	Approved
17	2	Graduate	Yes	500000	1600000	4	663	1300000	100000	1300000	700000	Approved
18	4	Not Graduate	Yes	2900000	9400000	14	780	2900000	2800000	6700000	4300000	Approved
19	2	Graduate	No	2700000	10300000	10	786	1000000	1000000	6200000	3300000	Approved
20	5	Graduate	No	6300000	14600000	12	652	10300000	3500000	21500000	5900000	Approved

First 20 rows of the dataset are being displayed. I chose not to display the row id as loan_id column already shows an id index.

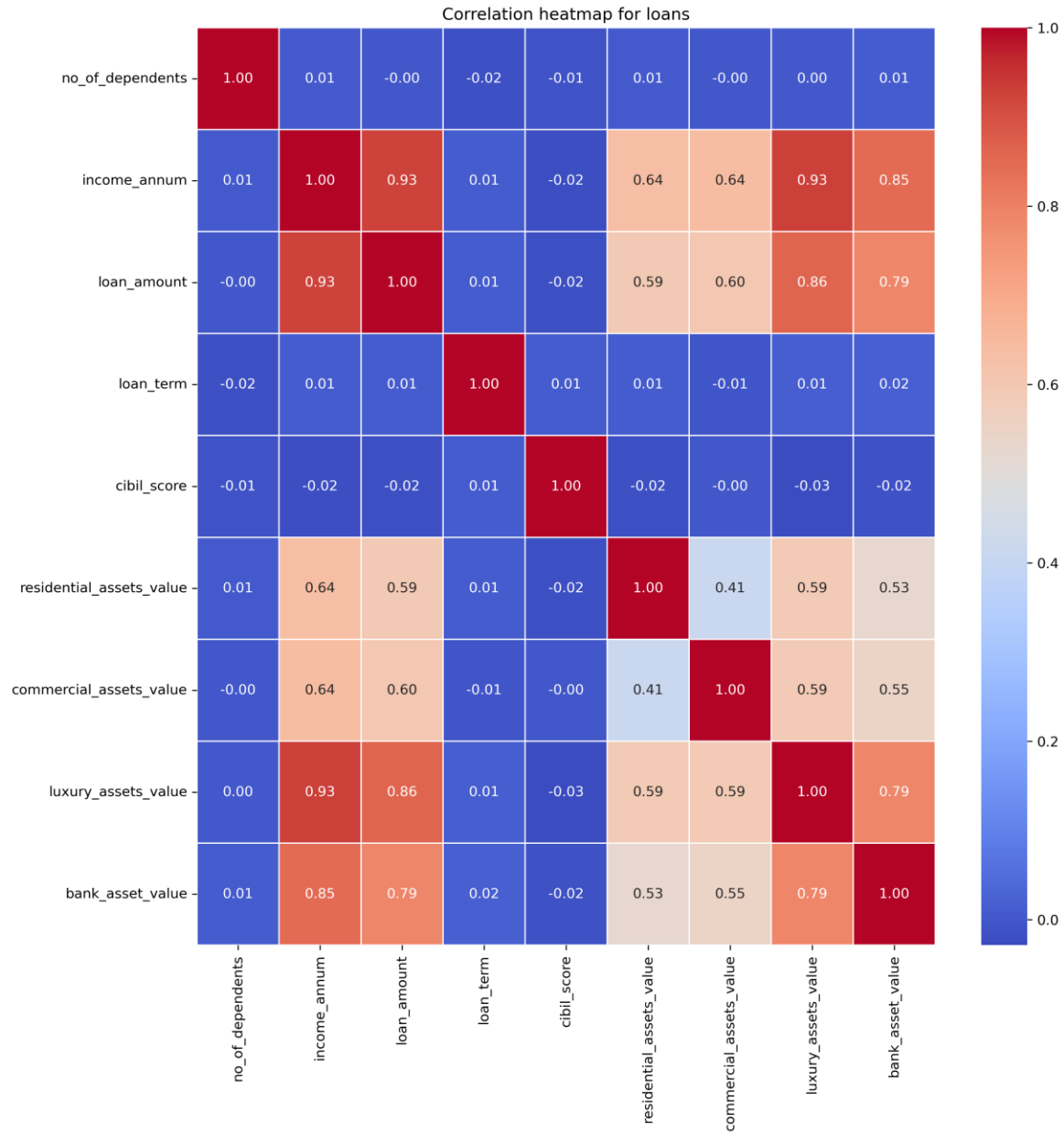
Also, the CSV had a leading space in every entry and feature column, so I removed it with .strip() because I could not get it to work with the heatmap plotting and analysis.

Task 3:

Feature	Important	In-Between	Not Important	Reason
loan_id	0	0	1	Loan identifier, no role.
dependents	0	1	0	More dependents may indicate greater financial responsibility
education	0	1	0	Education level may have relation with income and stability
self_employed	0	1	0	Employment type can affect income stability
income_annum	1	0	0	Higher income means greater ability to repay

loan_amount	1	0	0	Greater loan amount means greater repay obligations
loan_term	0	1	0	Repayment period may affect monthly payments
cibil_score	1	0	0	Credit score, direct indicator of finance
residential_assets_value	1	0	0	Valuable residential assets provide security
commercial_assets_value	1	0	0	Same as above
luxury_assets_value	0	1	0	May indicate wealth but not a direct measure of loan repay ability
bank_asset_value	1	0	0	Same as residential and commercial asset values

Task 4:



Heatmap generated by the Python program. Positive correlations are represented by warm colors, while negative correlations are represented by cool colors.

We can see that income has a high correlation with the assets owned, which is expected. Additionally, each asset feature has a high correlation with the rest of the asset features, meaning that someone who has the financial ability to own such assets is also likely to own assets of the other types as well.

Task 5:

```
loan_status_numeric    1.000000
cibil_score            0.770518
loan_amount            0.016150
commercial_assets_value 0.008246
bank_asset_value       -0.006778
residential_assets_value -0.014367
income_annum          -0.015189
luxury_assets_value    -0.015465
no_of_dependents       -0.018114
loan_term              -0.113036
Name: loan_status_numeric, dtype: float64
```

For Task 5, I converted loan_status to numeric (1 for accepted, 0 for rejected) and calculated the correlations with each feature. The closer to 1 each correlation is, the stronger the relationship is with approving a loan.

From the data above, we can conclude that a high CIBIL score has the highest correlation with loan approval, while all other factors have scores very close to 0 and thus play a very minor role, if any at all.

loan_status	Approved	Rejected
education		
Graduate	62.453358	37.546642
Not Graduate	61.976471	38.023529

Additionally, I chose education as the categorical feature to analyze the relation to loan_status. We can see from the data above that approval percentage for graduates is almost identical to that of non-graduates, meaning that there is not much of a difference between the two groups when approving a loan.

To conclude, the CIBIL score is very strongly associated with loan approval and education does not have a major effect when accepting or rejection loan applications.